

ARASH PARSA

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ABOUT ME

Scientist and data professional at the intersection of biology, computation, and applied machine learning. Experienced in building predictive models, analyzing complex datasets, and translating findings into actionable insights across healthcare, research, and industry settings. Background in bioengineering and data science with hands-on experience in clinical data, statistical analysis, and cross-functional collaboration.

EDUCATION

Georgia Institute of Technology

Masters of Science in Data Science/Analytics

Atlanta, GA August 2022 – July 2025

University of California, Santa Cruz

Bachelors of Science in Bioengineering/Bioinformatics

Santa Cruz, CA August 2015 – June 2021

EXPERIENCE

Machine Learning Research Collaborator – Cost-Aware Clinical Phenotyping

Dr. Vivek Rudrapatna's Clinical Data Science Group, UCSF

2026 – Present

San Francisco, CA

- Initiated the collaboration independently; scoped the cost bottleneck in EHR phenotyping directly with the physician-scientist stakeholder and proposed a confidence-gated cascade as the solution
- Architected a four-stage pipeline: (1) *Extract* – CPU BioBERT disease-NER; (2) *Retrieve* – entity-guided context selection; (3) *Cascade* – accept high-confidence entities on the cheap path, route low-confidence and zero-hit notes to a frontier LLM; (4) *Evaluate* – mention-level precision/recall/F1 against gold annotations with a token-cost proxy
- On an early pilot over the NCBI Disease corpus, escalating ~35% of notes raised precision from 0.62 to 0.75 while consuming only ~24% of the tokens a full-note frontier-LLM approach would require
- Engineered the system to run CPU-only with no GPU dependency and no LLM API required for the cheap path – a deliberate constraint that makes the approach deployable inside hospital firewalls where GPUs are unavailable
- Authored a literature review and an 11-slide novelty case establishing that individual cost levers (distillation, retrieval, cascading) are each proven, but none are combined into one confidence-gated cascade benchmarked across many phenotypes
- Obtained PhysioNet credentialing for MIMIC-III/IV; extending toward a weighted multi-dataset benchmark (MIMIC, n2c2) comparing distilled open-weight models against frontier models on joint accuracy-cost frontiers, with UCSF corpus validation and publication planned

Knowledge and Skills: Clinical NLP; named entity recognition; BioBERT; transformer fine-tuning; knowledge distillation; large language models; confidence-gated model cascades; retrieval/context selection; benchmark design; precision/recall/F1 evaluation; cost-accuracy tradeoff analysis; MIMIC-III/IV; n2c2; NCBI Disease; credentialed clinical data; CPU-constrained inference; PyTorch; HuggingFace Transformers; Python; Git; independent research scoping

Data Science Research Assistant in Clinical Research

In collaboration with Stanford-affiliated researchers

January 2026 – Present

San Francisco, CA

- Trichoscopic Oculomics – built deep learning classification pipeline integrating spatial transcriptomics and metabolic epidemiology features to reframe androgenetic alopecia as a systemic screening opportunity
- Endometriosis Severity Prediction – developed supervised ML models (random forest, gradient boosting, logistic regression) to phenotype and preoperatively predict endometriosis severity from clinical and demographic features extracted from EHR-derived datasets
- Performed end-to-end clinical data preprocessing: missing data imputation, feature encoding, class imbalance correction (SMOTE), and cross-validated model evaluation; presented results to clinical research stakeholders

Knowledge and Skills: Machine learning; clinical data analysis; EHR data; predictive modeling; phenotyping; Python; scikit-learn; TensorFlow; Pandas; data preprocessing; feature engineering; cross-validation; precision medicine; stakeholder communication

Machine Learning & Data Science Research Assistant

Washington Lab, UCSF

December 2025 – March 2026

San Francisco, CA

- Systematic review study: cataloged model architectures (CNNs, LSTMs, transformers), dataset characteristics, and performance benchmarks across 50+ studies on home video-based ML classification of autism and ADHD in pediatric populations
- Developed Python scripts to extract, standardize, and visualize study metrics (sensitivity, specificity, AUC) across heterogeneous reporting formats, enabling quantitative cross-study comparison for clinical decision support
- Manuscript to be submitted for publication in the near future

Knowledge and Skills: Systematic literature review; clinical data analytics; AI/ML in healthcare; model architecture analysis; performance benchmarking; Python; Pandas; Matplotlib; statistical analysis; data visualization

Machine Learning Scientist

Optelligence

September 2022 – Present Berkeley, CA

- Designed and trained supervised regression and classification models (XGBoost, neural networks) for performance prediction, cutting physical testing cycles by 50%
- Built end-to-end data preprocessing pipelines in Python (Pandas, NumPy, scikit-learn) including feature engineering, outlier detection, and normalization across heterogeneous datasets
- Constructed A/B testing framework with statistical significance testing (t-tests, bootstrap CIs) to evaluate model variants, improving deployment efficiency by 25%
- Wrote SQL queries to extract and join data from PostgreSQL databases; created data visualizations and dashboards to communicate findings to cross-functional stakeholders

Knowledge and Skills: Python; SQL; PostgreSQL; TensorFlow; PyTorch; scikit-learn; XGBoost; Pandas; NumPy; feature engineering; statistical analysis; A/B testing; dashboard development; data visualization; stakeholder communication

R & D Engineer

Amgen & Thermofisher

August 2021 – May 2022 Thousand Oaks, CA & Pleasanton, CA

- Automated quality control workflows using Python and SQL, reducing manual review time and maintaining 99.8% assay accuracy across 100+ daily COVID-19 diagnostic samples
- Developed statistical process control (SPC) dashboards to monitor HPLC/UPLC instrument drift, enabling early anomaly detection and reducing reagent waste by \$50K annually
- Performed multivariate data analysis on chromatography outputs using R and Excel; identified systematic variance sources and presented findings and recommendations to cross-functional clinical and laboratory stakeholders

Knowledge and Skills: Python; R; SQL; Excel; statistical process control; dashboard development; quality assurance; chromatography data analysis; multivariate analysis; anomaly detection; clinical laboratory workflows; stakeholder communication

Research Assistant – Progesterone Biosynthesis (iGEM)

University of California, Santa Cruz

2018

Santa Cruz, CA

- Engineered *Y. lipolytica* for progesterone biosynthesis via metabolic pathway engineering; designed 5-gene cassette for heterologous expression
- Executed 3 parallel cloning strategies: Gibson Assembly, yeast-mediated cloning, and Cre-Lox recombination; validated constructs with gel electrophoresis, colony PCR, and Sanger sequencing
- Led international outreach across 15+ countries; co-authored 60-page published research report

Knowledge and Skills: Metabolic pathway engineering; gene cassette design; Gibson Assembly; yeast transformation; Cre-Lox recombination; plasmid cloning; PCR; gel electrophoresis; Sanger sequencing; bioinformatics; primer design

Scientific Researcher & Research Consultant

Apex Energetics

January 2016 – November 2019

Irvine, CA

- Designed and maintained a structured literature database of 200+ papers/month on 15+ nutraceutical compounds; applied NLP-assisted tagging to surface trends in clinical efficacy data
- Synthesized evidence across heterogeneous study designs (RCTs, meta-analyses, in vitro) to produce quantitative summary reports, saving 20 hours/week of manual review for research leads

Knowledge and Skills: NLP-assisted literature review; evidence synthesis; RCTs; meta-analyses; clinical efficacy data; quantitative reporting; database design and maintenance

PERSONAL PROJECTS

ICU Readmission Prediction from Clinical EHR Data (MIMIC-IV)

Personal Project – GitHub

2025 – 2026

Berkeley, CA

- Built a 30-day ICU readmission prediction model using MIMIC-IV, a publicly available critical care dataset of 300K+ hospital admissions from Beth Israel Deaconess Medical Center
- Wrote complex SQL queries in PostgreSQL to extract and join patient demographics, diagnoses (ICD codes), lab results, vitals, and medication records across multiple relational tables; engineered 40+ clinical features
- Trained and compared logistic regression, random forest, and XGBoost classifiers; performed hyperparameter tuning via cross-validation, achieving AUC of 0.78 on held-out test set
- Handled class imbalance with SMOTE and threshold tuning; generated SHAP explanations to identify top predictive features (e.g., length of stay, lactate levels, comorbidity index) for clinical interpretability
- Visualized model performance (ROC curves, calibration plots, feature importance) and presented findings in a reproducible Jupyter notebook with clinical context documentation

Knowledge and Skills: Python; SQL; PostgreSQL; scikit-learn; XGBoost; Pandas; SHAP; EHR data; ICD codes; clinical feature engineering; SMOTE; cross-validation; predictive modeling; data visualization; MIMIC-IV

Multi-Class Text Classification with Traditional ML and Transformers

Personal Project – GitHub

2025 – 2026

Berkeley, CA

- Built an end-to-end NLP pipeline for multi-class news topic classification on the AG News dataset (120K articles, 4 categories), comparing TF-IDF + logistic regression, word embeddings + LSTM, and fine-tuned DistilBERT
- Implemented text preprocessing pipeline: tokenization, stopword removal, lemmatization, and vocabulary analysis; explored feature representations from bag-of-words to contextual embeddings
- Fine-tuned DistilBERT using HuggingFace Transformers, achieving 94% accuracy vs. 91% for TF-IDF baseline; analyzed misclassification patterns across categories with confusion matrices and per-class F1 scores
- Deployed the best model as a REST API using FastAPI with a simple web interface for real-time text classification; containerized with Docker for reproducibility

Knowledge and Skills: Python; NLP; scikit-learn; PyTorch; HuggingFace Transformers; TF-IDF; BERT; text classification; FastAPI; Docker; model deployment; data visualization

PUBLICATIONS

Machine Learning-Based Phenotyping and Preoperative Prediction of Endometriosis Severity in Patients with Unexplained Infertility

Abstract submitted to Stanford AI in Medicine and Medical Imaging (AIMI) Conference. Manuscript in preparation for publication within Journal of Clinical Medicine.

Trichoscopic Oculomics: Integrating Machine Learning, Spatial Transcriptomics, and Metabolic Epidemiology to Reframe Androgenetic Alopecia as a Systemic Screening Opportunity

Manuscript in preparation for publication.

A Systematic Review of Home Video-Based Machine Learning Classification of Autism and ADHD in Pediatric Populations

Manuscript in preparation for publication.

25-min carotenoid blood test for lung cancer via a triple photonic resonance

Manuscript in preparation for publication.

TECHNICAL SKILLS

Programming: Python, SQL, R, MATLAB, C++, JavaScript

Machine Learning & AI: scikit-learn, XGBoost, TensorFlow, PyTorch, HuggingFace Transformers, LLMs, classification, regression, clustering, NLP, deep learning, predictive modeling

Health Data & Informatics: EHR data analysis, clinical NLP, clinical data preprocessing, patient phenotyping, clinical decision support, health data standards

Data Science & Statistics: Statistical modeling, hypothesis testing, A/B testing, experimental design, feature engineering, time series analysis

Data Analysis & Visualization: Pandas, NumPy, Matplotlib, Seaborn, PostgreSQL, Excel, dashboard development

Bioinformatics: KEGG, sequence alignment, plasmid design, genome analysis, primer design

Molecular Biology: PCR, Gibson Assembly, yeast transformation, Cre-Lox, gel electrophoresis, Sanger sequencing

Cloud & Infrastructure: AWS, Docker, Git, REST APIs, CI/CD